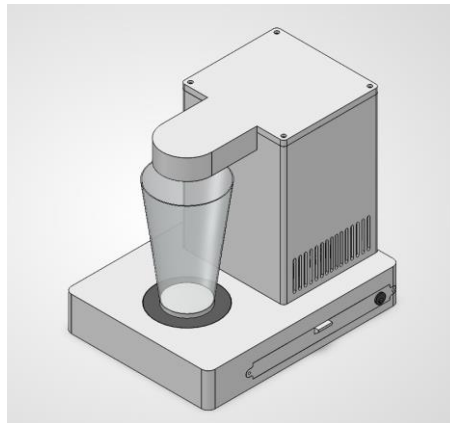




DiyaMithuru

BUSINESS MODEL, PRODUCT COST & SWOT ANALYSIS



SMART LIQUID FILLING ASSISTANT

DiyaMithuru is an inclusive smart liquid filling assistant designed to make routine liquid dispensing safer and easier. The primary design focus is visually impaired users, while the commercial concept also considers older adults, caregivers, children/baby-care routines, and people who need support with regular hydration.

The proposed business approach is a direct-to-customer model supported by institutional partnerships, a simple web-based pre-order channel, an upfront payment option, and a 0% three-month installment plan. The standard selling price is LKR 15,000, with a limited promotional cash price of LKR 14,250.

1. BUSINESS MODEL

The business model converts the engineering prototype into a practical customer offering. It focuses on affordability, accessibility, local support, and a clear route from prototype validation to small-batch sales.

Customer Segments Primary: visually impaired individuals. Secondary: older adults, caregivers, families, rehabilitation users and people who need assisted hydration. Institutional: hospitals, elder-care facilities, rehabilitation centres, NGOs and accessibility programmes.	Value Proposition A simple smart filling assistant that reduces manual pouring, supports safer liquid handling, improves independence and provides an accessible home-use experience at a locally targeted price.
Channels Direct website/pre-order page, social media promotion, university exhibitions, demonstrations, healthcare/rehabilitation partnerships, NGOs and direct institutional sales.	Customer Relationships Product demonstration, personal onboarding, one-year hardware warranty, user guidance, after-sales support and feedback collection for future product improvement.
Revenue Streams Standard outright sale: LKR 15,000. Promotional full-payment price: LKR 14,250 (5% discount). 0% installment plan: LKR 5,000 per month for 3 months. Future revenue may include institutional/bulk sales and upgraded product versions.	Key Activities Prototype refinement, assembly, quality checking, user testing, supplier sourcing, marketing, order fulfilment, customer onboarding, warranty support and continuous design improvement.
Key Resources ESP32-based control system, sensors and pump assembly, product enclosure, design files, website/pre-order channel, engineering know-how, user feedback and supplier network.	Key Partners Electronics suppliers, fabrication/enclosure providers, healthcare/rehabilitation organisations, disability-support organisations, delivery partners and academic/technical mentors.
Cost Structure Electronic components, enclosure/fabrication, pump and tubing, power adapter, assembly/testing labour, packaging, delivery, warranty reserve, marketing and website/administration costs.	Growth Path Start with small-batch direct sales, validate reliability and user satisfaction, build institutional partnerships, improve manufacturability, then expand to wider assistive/home-care markets.

Business model statement: DiyaMithuru will initially follow a direct-to-customer, small-batch model to keep the product affordable and maintain close contact with users. Institutional partnerships can later support bulk deployment and broader market reach.

1.1 Proposed Sales and Payment Model

Option	Customer Payment	Total Revenue	Purpose
Standard full payment	LKR 15,000 once	LKR 15,000	Normal retail sale
Promotional full payment	LKR 14,250 once	LKR 14,250	5% launch promotion
0% installment plan	LKR 5,000 x 3 months	LKR 15,000	Improves affordability / access

2. PRODUCT COST AND PRICING

The following cost model is an estimated planning budget for one DiyaMithuru unit. It is intended for the assignment business case and should be updated using actual purchase receipts, supplier quotations and fabrication costs before the final presentation.

Item	Qty	Estimated Unit Cost (LKR)	Estimated Total (LKR)
ESP32 controller	1	1,600	1,600
Ultrasonic cup-detection sensor	1	450	450
Temperature sensor	1	500	500
1-channel relay module	1	250	250
Logic-level shifter	1	300	300
DC liquid pump	1	1,200	1,200
Wall power adapter / power unit	1	1,000	1,000
Tubing and fluid connectors	1	400	400
Product enclosure / fabricated body	1	1,700	1,700
Wiring, PCB/prototyping board, fasteners	1	600	600
Audio indication hardware	1	450	450

Packaging, user guide and warranty materials	1	350	350
Assembly and functional testing allowance	1	700	700
Estimated manufacturing cost per unit			LKR 9,500

2.1 Pricing and Margin

Scenario	Selling Price	Gross Profit / Unit	Gross Margin
Standard retail	LKR 15,000	LKR 5,500	36.7%
5% promotional full payment	LKR 14,250	LKR 4,750	33.3%
Installment (3 x LKR 5,000)	LKR 15,000	LKR 5,500	36.7%

Gross margin is before delivery, warranty claims, marketing, payment collection costs, taxes/fees and other overheads. These should be tracked separately during commercialization.

2.2 Small-Batch Financial Projection

Scenario	20-Unit Revenue	20-Unit Product Cost	Gross Contribution
Standard price	LKR 300,000	LKR 190,000	LKR 110,000
All units at promo price	LKR 285,000	LKR 190,000	LKR 95,000
Illustrative mixed sales (10 standard + 10 promo)	LKR 292,500	LKR 190,000	LKR 102,500

2.3 Financial Plan for the First Product

- Use the website/pre-order QR code and product demonstration to collect the first confirmed order.
- Offer the standard price of LKR 15,000, or a limited 5% promotional full-payment price of LKR 14,250.
- Offer the 0% installment option at LKR 5,000 per month for three months while keeping the total price at LKR 15,000.
- Purchase components against confirmed orders where possible to reduce inventory risk and preserve working capital.
- Record actual component, fabrication, delivery and warranty costs for each unit so the final selling price can be refined using real data.

Estimated first-unit economics at promotional price: Revenue LKR 14,250 - estimated product cost LKR 9,500 = estimated gross contribution LKR 4,750 before delivery, warranty and marketing overheads.

3. SWOT ANALYSIS

The SWOT analysis identifies internal product/business factors and external market factors that can affect DiyaMithuru during prototype validation and early commercialization.

STRENGTHS

- Inclusive, user-centered concept with a clear primary accessibility need.
- Locally developed solution that can be supported and adapted for Sri Lankan users.
- Simple embedded architecture based on commonly available components.
- Automatic cup detection, liquid dispensing, temperature monitoring and audio indication improve ease of use.
- Potential to serve more than one customer segment while retaining a strong accessibility focus.
- Web-based marketing/pre-order channel enables low-cost direct customer reach.

WEAKNESSES

- Alpha-stage product with limited long-term reliability and field testing.
- Current unit cost may be high relative to mass-produced household dispensers.
- Performance depends on correct sensor positioning, calibration and consistent cup placement.
- Small-batch fabrication can cause variation in enclosure quality and assembly time.
- Warranty, delivery and installment commitments can create extra cost and cash-flow pressure.
- Product is not a medical measuring device; claims about patient fluid requirements must remain carefully limited.

OPPORTUNITIES

- Growing interest in inclusive design, independent living and smart-home assistance.
- Partnership opportunities with hospitals, rehabilitation centres, elder-care services and NGOs.
- Potential institutional/bulk orders after successful user validation.
- Future versions can improve portability, manufacturability, connectivity and personalised accessibility features.
- Local manufacturing and assembly may reduce import dependence and improve after-sales support.
- Positive university exhibition and survey feedback can support further product development.

THREATS

- Competition from low-cost water dispensers, smart appliances and imported assistive products.
- Component price changes or shortages can reduce margins.
- Electrical/liquid safety concerns may require stronger testing and compliance before commercial release.
- Customer adoption may be slow without demonstrations and trust-building.
- Sensor or pump failure can directly affect customer confidence in the product.
- Unvalidated marketing claims could create reputational or regulatory risk.

3.1 Actions from the SWOT

Priority	Action
Validate reliability	Run repeat filling tests, temperature-sensor checks and user trials; document failures and improvements.
Control product cost	Use supplier quotations, standardise components and redesign the enclosure for easier small-batch manufacturing.
Build customer trust	Demonstrate the product, clearly explain limitations, provide onboarding and publish genuine user feedback.
Manage commercial risk	Define warranty terms, installment collection rules, delivery cost policy and a small warranty reserve per unit.

4. CONCLUSION

DiyaMithuru has a credible early-stage business pathway because the product addresses an identifiable accessibility need, uses a comparatively simple embedded architecture and can initially be sold through direct customer relationships. The proposed LKR 15,000 standard price and LKR 14,250 promotional price provide a workable starting point under the estimated LKR 9,500 unit-cost model, but the financial case must be updated with real supplier and fabrication data before commercial launch.

The immediate business priority is not large-scale expansion. It is to validate reliability, confirm customer willingness to pay, control unit cost and build trust through real demonstrations and user feedback. These results can then support institutional partnerships and future product development.